

Active Directory Home Lab

Windows Server 2022 Domain Administration Project

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1. Project Overview

I built this project to learn how Windows domain is managed in a business environment. I created a small Active Directory lab using Windows Server 2022 and a Windows 11 client computer. The project gave me hands-on practice with user and group management, Group Policy, shared folders, permissions, PowerShell automation, security auditing, and troubleshooting.

2. Project Objectives

The main goals of this project were to:

- Build and configure a Windows Server domain.
- Create an organized Active Directory structure for different departments.
- Manage users and access through security groups.
- Join a Windows client computer to the domain.
- Apply security settings through Group Policy.
- Configure department-based file sharing and permissions.
- Automate user onboarding with PowerShell.
- Verify successful and failed logon activity through Event Viewer.
- Practice troubleshooting issues within a virtual lab environment.

3. Lab Environment

Component	Configuration
Hypervisor	Oracle VirtualBox
Domain Controller	DC01
Client Computer	CLIENT-01
Server OS	Windows Server 2022 Evaluation
Client OS	Windows 11
Domain Name	NOLANCYBER.local
DNS Server	DC01
Active Directory	Active Directory Domain Services (AD DS)
PowerShell	User for user provisioning and automation

The lab environment was built using two virtual machines in Oracle VirtualBox. One virtual machine was configured as the Windows Server domain controller, while the second served as a Windows 11 client joined to the domain. This setup allowed me to practice common administrative tasks in a controlled environment without requiring physical hardware

4. Technologies Used

Operating Systems

- Windows Server 2022 Evaluation
- Windows 11

Virtualization

- Oracle VirtualBox

Windows Server Roles and Features

- Active Directory Domain Services (AD DS)
- Domain Name System (DNS)
- Group Policy Management

Administration Tools

- Active Directory Users and Computers (ADUC)
- Group Policy Management Console (GPMC)
- Event Viewer
- PowerShell

Security & Access Management

- Organizational Units (Ous)
- Security Groups
- NTFS Permissions
- Shared Folder Permissions
- Windows Security Auditing

Automation

- PowerShell Scripting
- CSV user provisioning (Import-Csv)

Networking

- TCP/IP
- Static IP Address Configuration
- Domain Name Resolution (DNS)

5. Network and Domain Configuration

Setting	Configuration
Domain Name	NOLANCYBER.local
Domain Controller	DC01
Client Computer	CLIENT-01
DC01 IP Address	192.168.1.10
CLIENT-01 IP Address	192.168.1.20
DNS Server	192.168.1.10
Network Type	Internal VirtualBox Network

The first step of the project was creating a Windows domain using Windows Server 2022. After installing Active Directory Domain Services and DNS on the server, I promoted the server to a domain controller and created the **NOLANCYBER.local** domain.

Once the domain was running, I configured a Windows 11 virtual machine with a static IP address and pointed its DNS server to the domain controller. After verifying network connectivity, I joined the client computer to the domain and confirmed that domain users could successfully log in.

6. Active Directory Structure

To keep the domain organized, I created a structured Active Directory environment using Organizational Units (OUs), security groups, and user accounts for each department. Organizing resources this way made it easier to manage permissions, apply Group Policy, and simulate how users are typically grouped in a business environment.

Organizational Units (Ous)

I organized the domain into separate Organizational Units for each department instead of placing every user in the default Users container. This structure makes administration more

organized and allows policies and permissions to be managed by department rather than by individual user accounts.

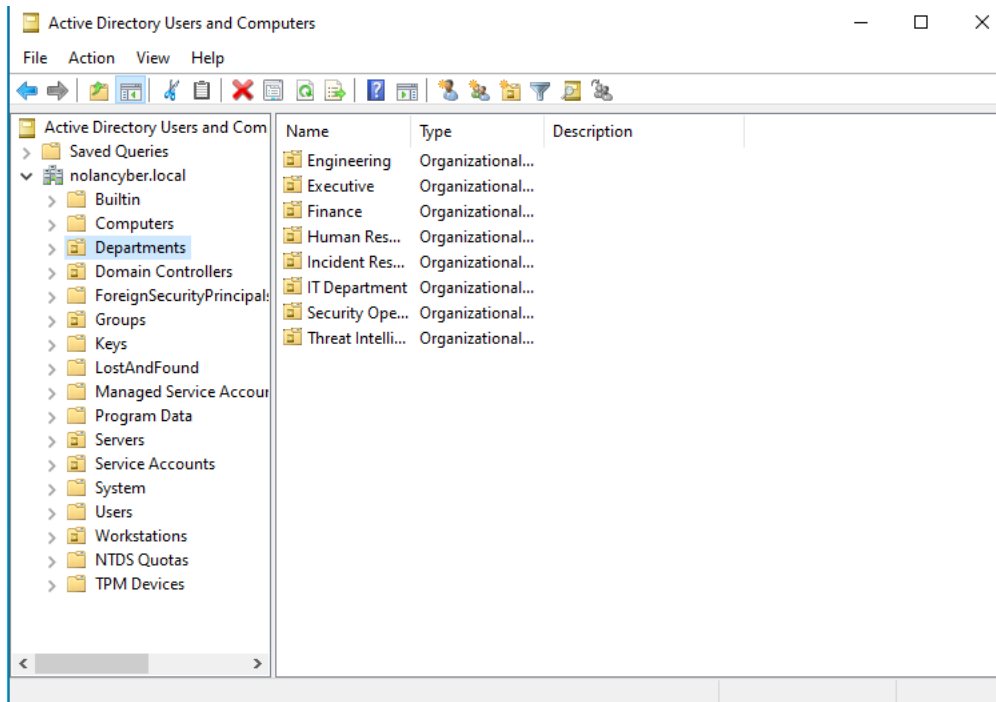


Figure 1. Active Directory Organizational Unit structure used to organize departments and simplify administration.

Security Groups

To simplify permission management, I created security groups for each department rather than assigning permissions to individual users. This approach makes it easier to manage access because permissions can be assigned to a group instead of being updated for every user account individually. If a new employee joins a department, they can simply be added to the appropriate security group.

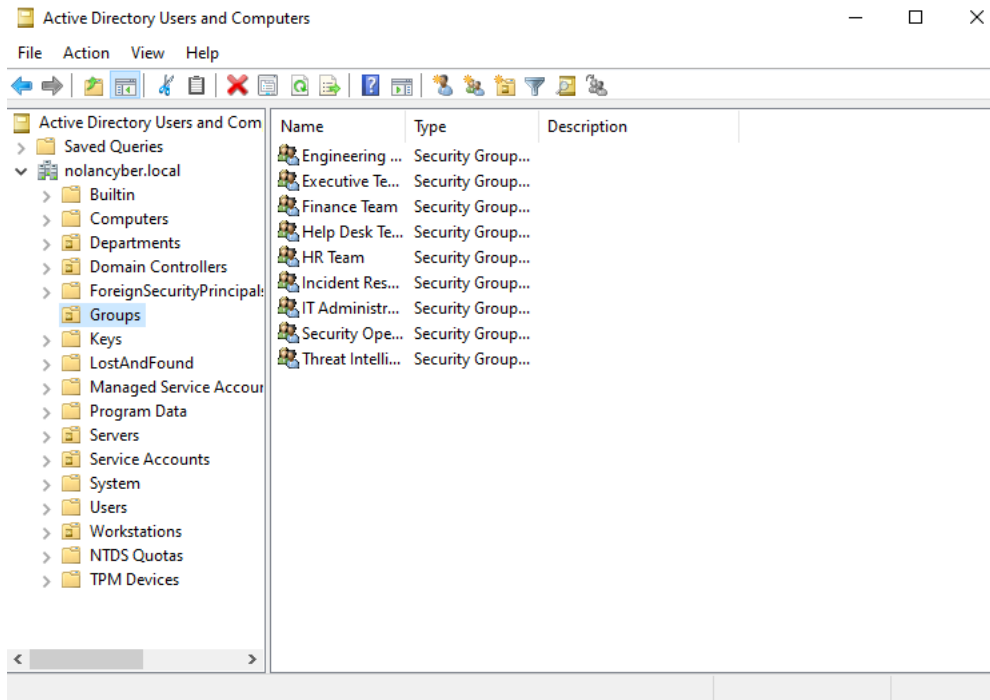


Figure 2. Department-based security groups created to simplify permission management and support role-based access control.

User Accounts

User accounts were created for each department to simulate employees in a small business environment. Each account was placed in the appropriate Organizational Unit and assigned to the correct security group based on its role. This organization made it easier to manage permissions and test access to shared folders throughout the project.

To keep the environment consistent, I followed a simple username naming convention using the employee's first initial and last name (for example, **mcurie** and **alennox**). Later in the project, I also automated user creation with PowerShell by importing employee information from a CSV file, allowing multiple accounts to be created and assigned to the correct Organizational Units and security groups.

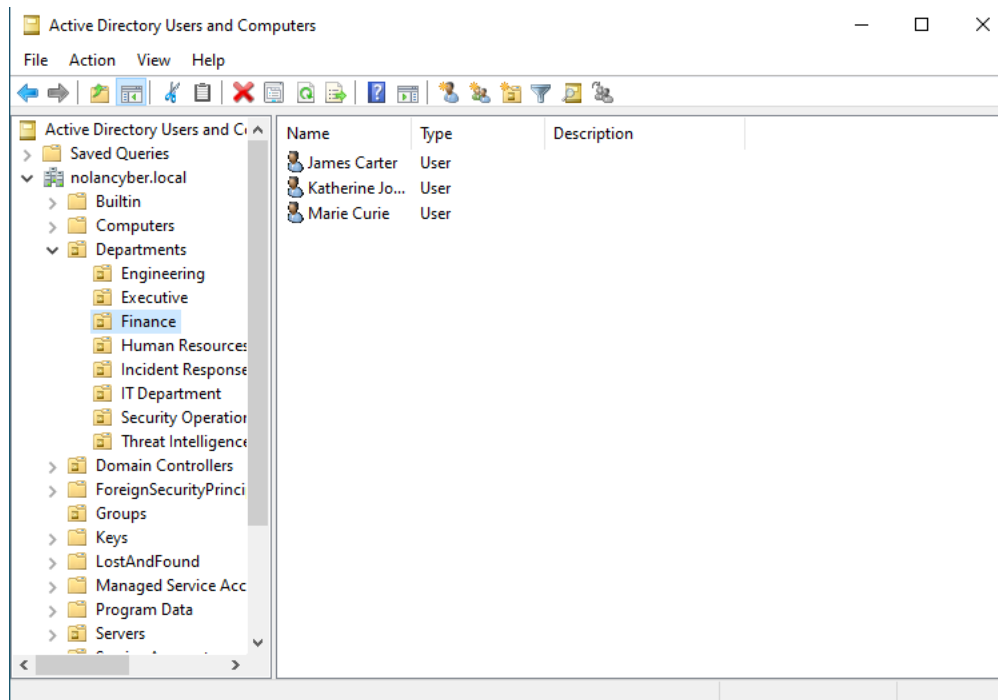


Figure 3. User accounts organized by department using a consistent naming convention and group membership to support centralized administration.

7. Group Policy Configuration

Group Policy was used to apply security settings across the domain and provide a more realistic administrative environment. Instead of configuring each computer or user individually, policies were applied centrally through Group Policy Management. This allowed security settings to be managed consistently while demonstrating how administrators enforce standards within a Windows domain.

Password and Account Policies

Password and account policies were configured to improve the security of the domain. These settings included password complexity requirements, password history, maximum password age, and account lockout settings to help protect user accounts from unauthorized access and repeated failed sign-in attempts.

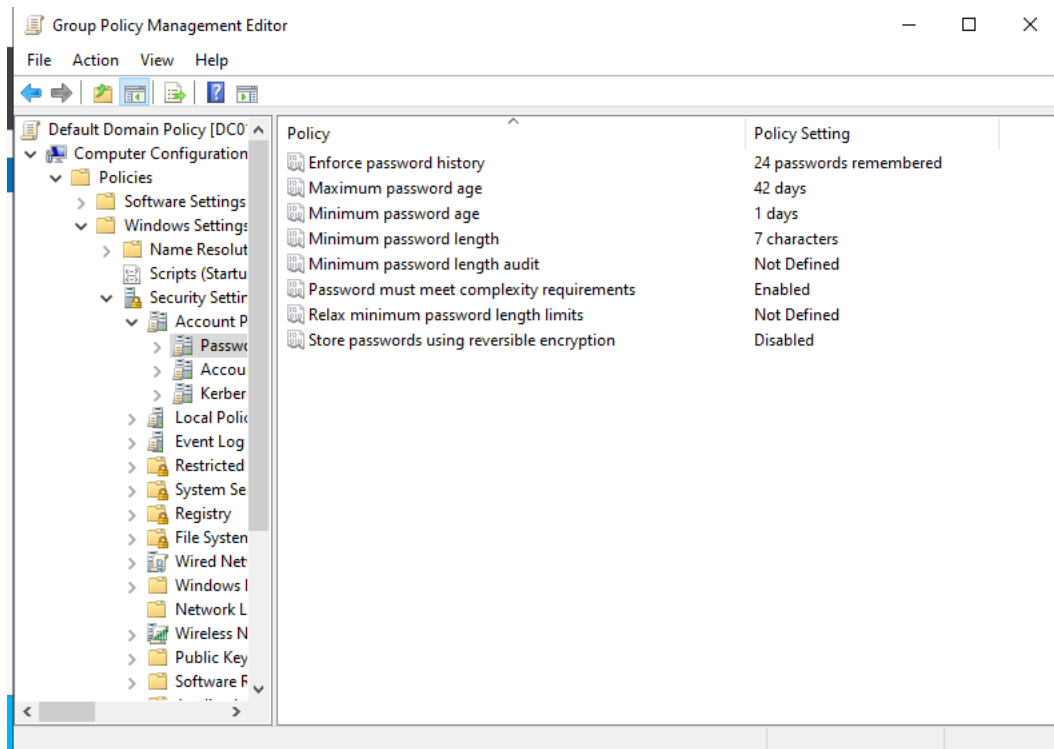


Figure 4. Password and account policies configured through Group Policy to improve domain security.

Security Baseline

A security baseline Group Policy Object was applied to strengthen the overall security of the lab environment. This provided additional security settings beyond the default configuration and helped simulate the types of policies commonly applied in business environments.

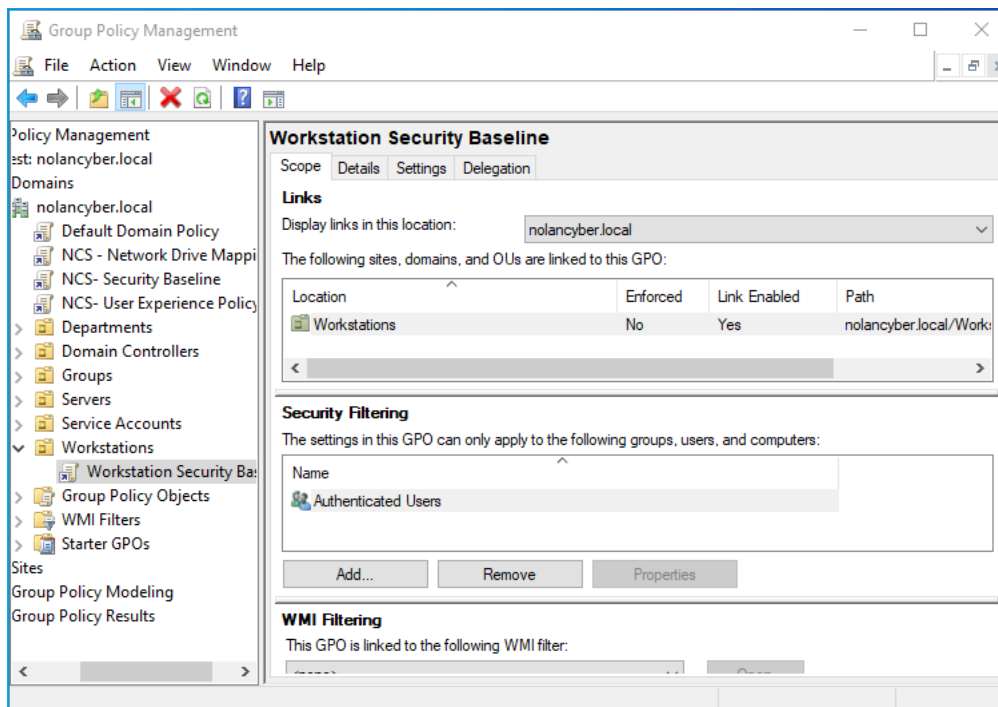


Figure 5. Security baseline Group Policy Object applied to strengthen workstation security.

Interactive Logon Banner

An interactive logon banner was configured to display a legal notice before users signed in. This type of policy is commonly used in organizations to notify users that systems are intended for authorized use and that activity may be monitored.

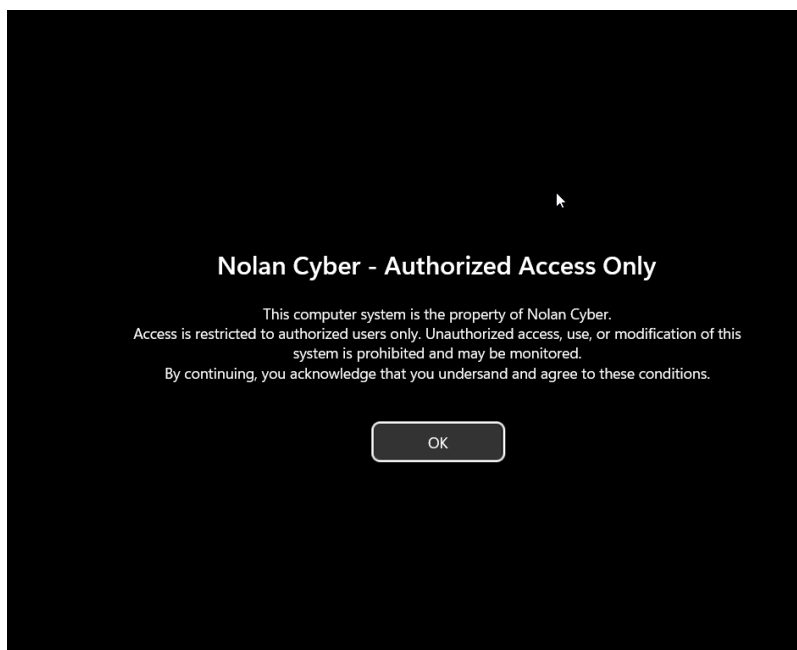


Figure 6. Interactive logon banner displayed before user authentication.

8. File Sharing and Permissions

Shared folders were created to simulate department file storage within the domain. Access was controlled using Active Directory security groups rather than assigning permissions to individual users. This approach simplifies administration because permissions can be managed by department, making it easier to add or remove users without changing folder permissions individually.

Shared Folder Configuration

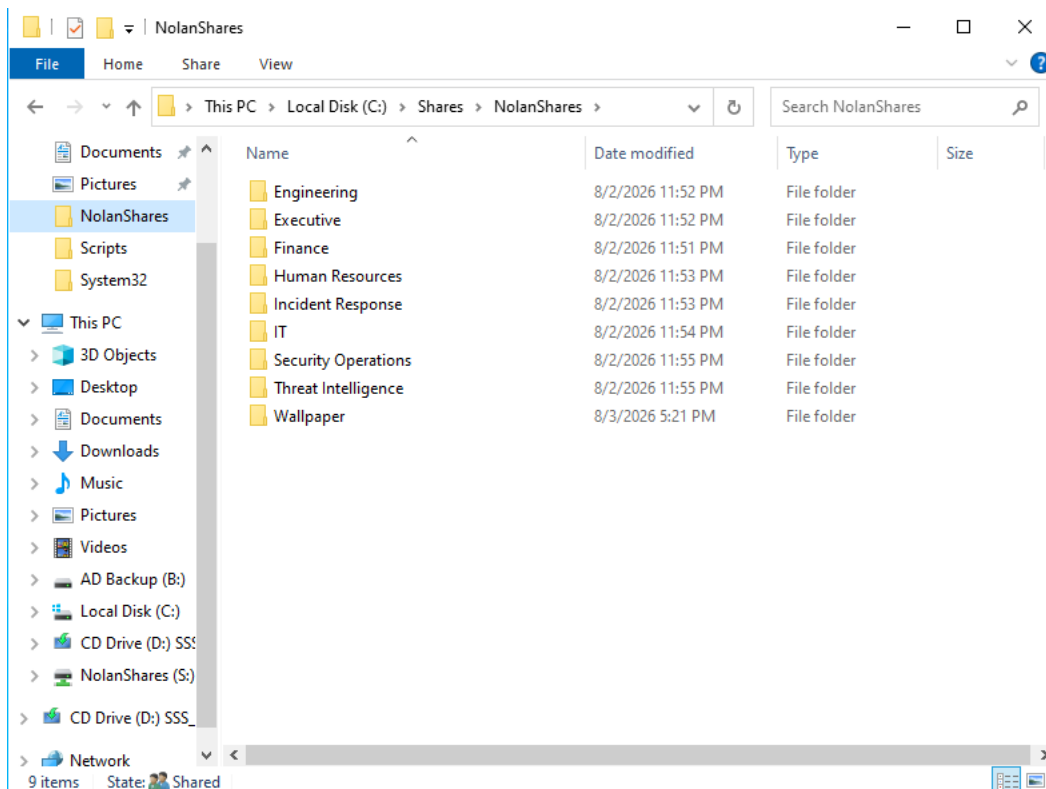


Figure 7. Department folders created on the shared file server for role-based access.

Permission Testing

After configuring permissions, I tested access by signing in as users from different departments. Each user was able to access their assigned department folder while being restricted from folders belonging to other departments. This confirmed that the security groups and folder permissions were working as intended.

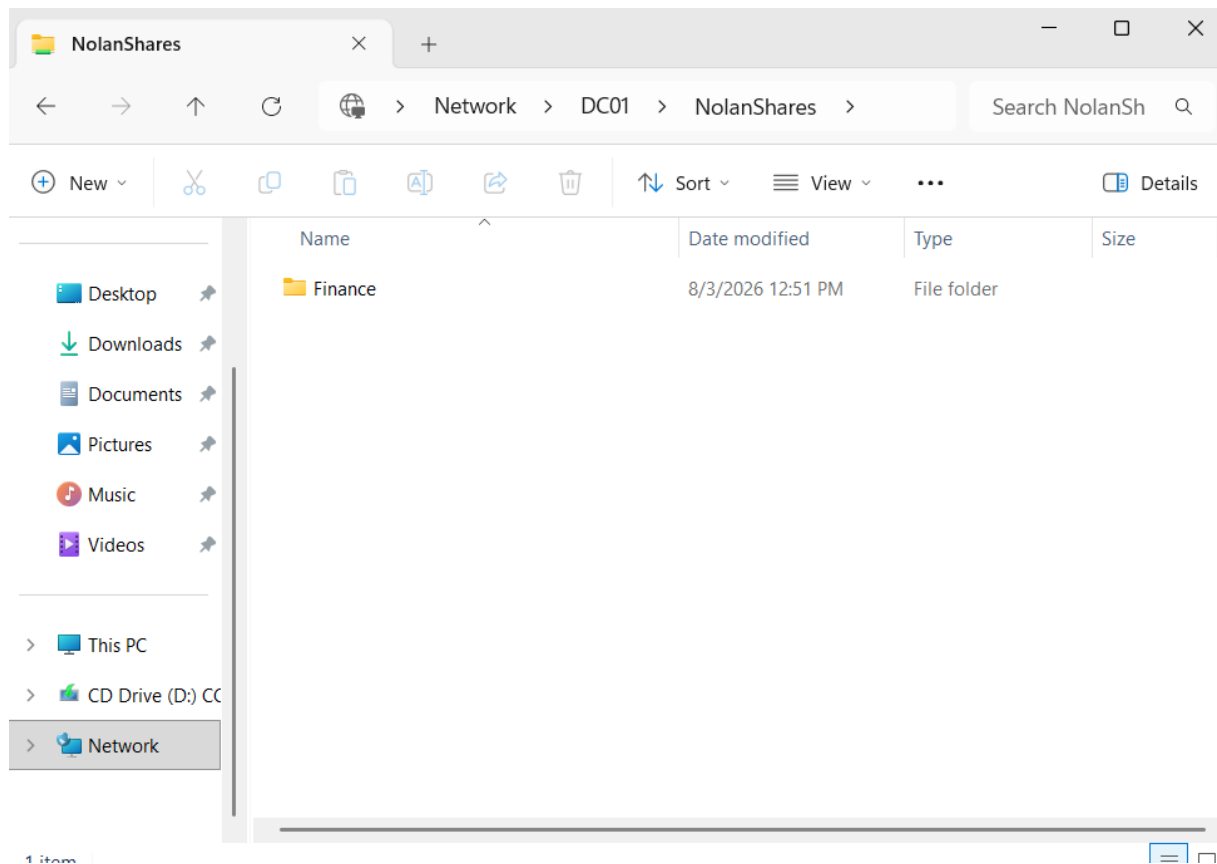


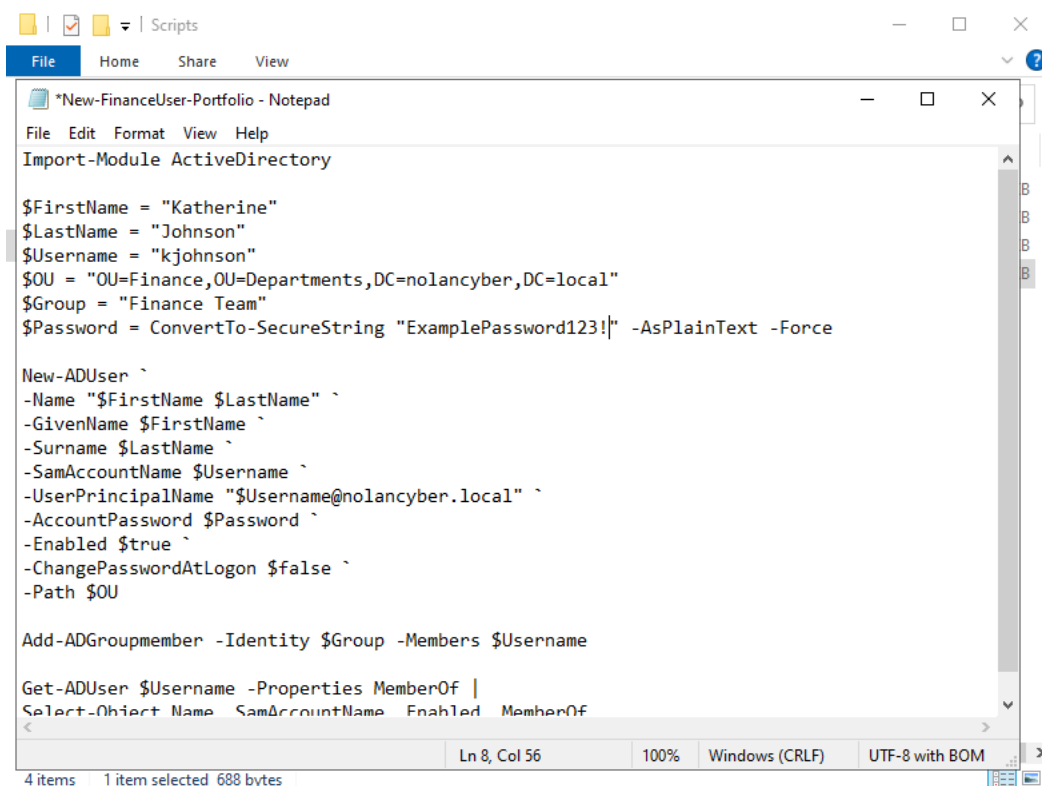
Figure 8. Departmental permission testing confirmed that the Finance user could access and create files within the assigned Finance resource.

9. PowerShell Automation

PowerShell was used to automate user account creation and reduce repetitive administrative tasks. Instead of creating every user manually, I developed scripts that created accounts, assigned them to the correct Organizational Unit, and added them to the appropriate security group. This demonstrated how automation can improve consistency and efficiency when managing Active Directory.

Single User Provisioning

To better understand how PowerShell interacts with Active Directory, I first created a script that provisions a single user account. This helped me learn the required Active Directory cmdlets and verify that users could be created, placed in the correct Organizational Unit, and assigned to the proper security group before moving on to bulk automation.



```
File Edit Format View Help
Import-Module ActiveDirectory

$FirstName = "Katherine"
$LastName = "Johnson"
$Username = "kjohnson"
$OU = "OU=Finance,OU=Departments,DC=nolancyber,DC=local"
$Group = "Finance Team"
$Password = ConvertTo-SecureString "ExamplePassword123!|" -AsPlainText -Force

New-ADUser `
-Name "$FirstName $LastName" `
-GivenName $FirstName `
-Surname $LastName `
-SamAccountName $Username `
-UserPrincipalName "$Username@nolancyber.local" `
-AccountPassword $Password `
-Enabled $true `
-ChangePasswordAtLogon $false `
-Path $OU

Add-ADGroupmember -Identity $Group -Members $Username

Get-ADUser $Username -Properties MemberOf |
Select-Object Name SamAccountName Enabled MemberOf
```

Ln 8, Col 56 100% Windows (CRLF) UTF-8 with BOM

4 items 1 item selected 688 bytes

Figure 9. PowerShell script used to automate creation of a single Active Directory user.

Bulk User Provisioning

After verifying the single-user script, I expanded the automation by importing employee information from a CSV file. The script automatically created multiple user accounts, placed them into the correct Organizational Units, and added each user to the appropriate security group based on department information contained in the CSV file.

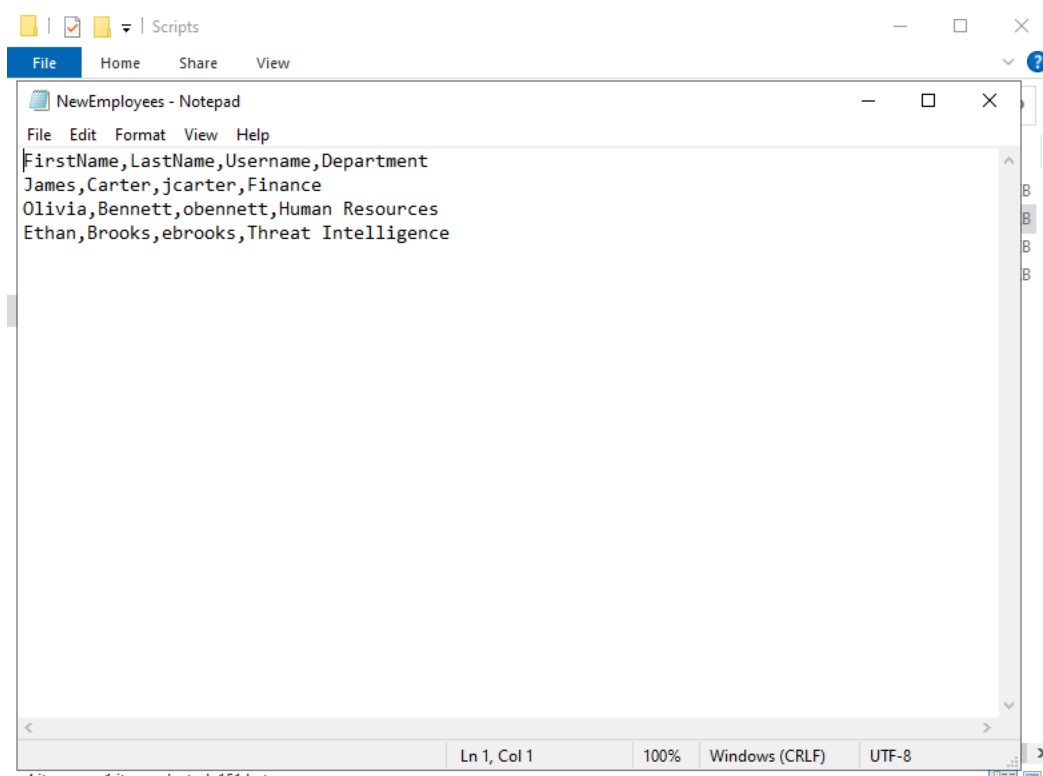


Figure 10. CSV file containing employee information used for automated account creation.

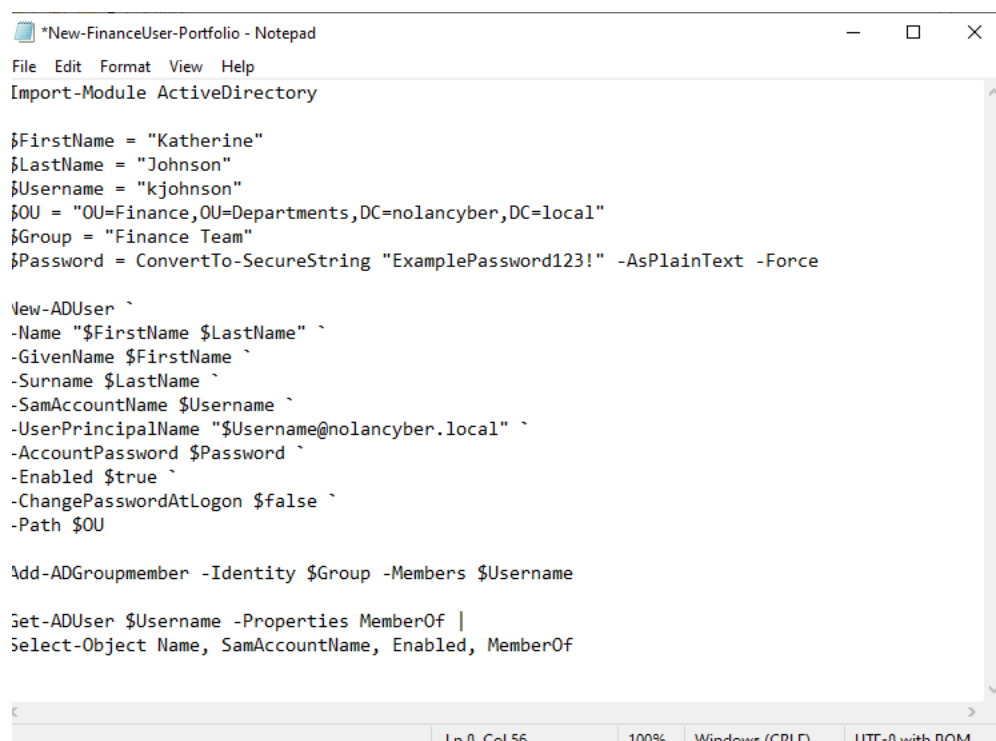


Figure 11. PowerShell script used to automate bulk user creation from a CSV file

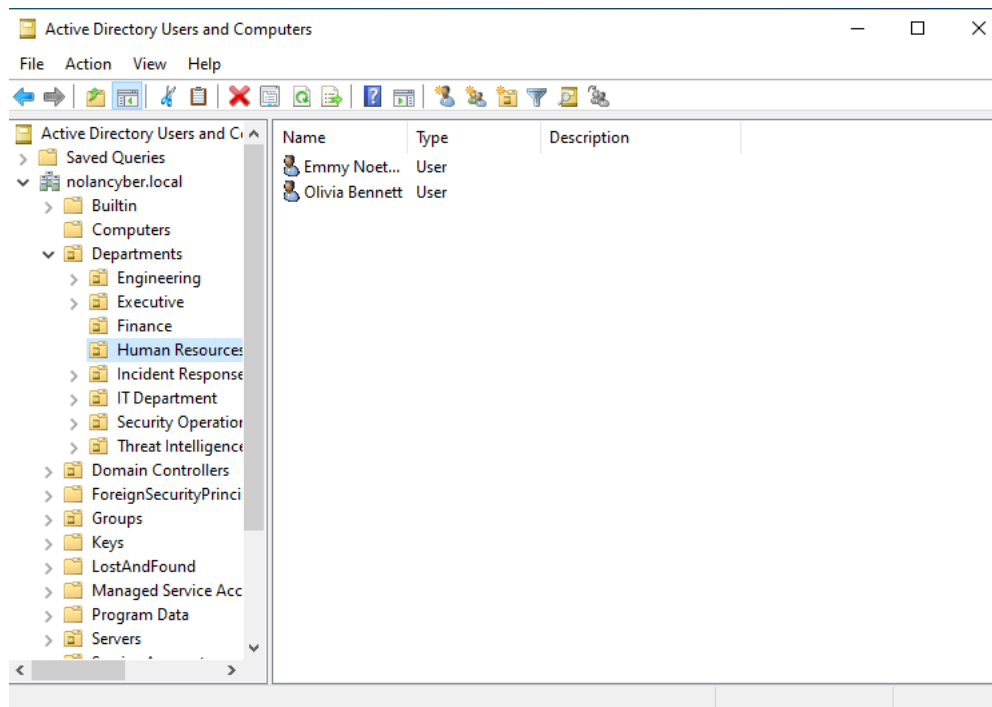


Figure 12. User accounts successfully created through PowerShell automation.

10. Security Auditing

After configuring the domain, I enabled and verified Windows Security Auditing to monitor authentication activity. Event Viewer was used to confirm that successful and failed logon attempts were being recorded correctly. Reviewing these events helped me understand how Windows tracks authentication and how administrators can use audit logs when troubleshooting or investigating security issues.

Successful Logon Events

Successful logons were verified by signing in with domain user accounts and reviewing Event ID 4624 in Event Viewer. These events confirmed that authentication requests were successfully processed by the domain.

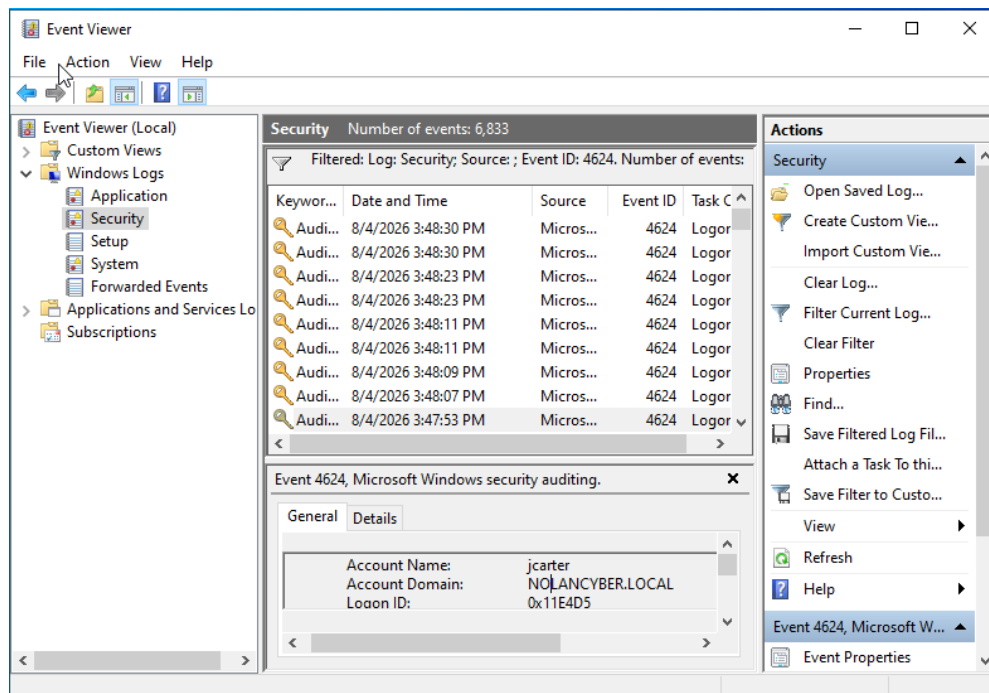


Figure 13. Event ID 4624 confirming a successful domain logon.

Failed Logon Events

Failed logons were tested by intentionally entering an incorrect password for a domain user account. Event ID 4625 was recorded in Event Viewer, confirming that failed authentication attempts were being audited correctly. This demonstrated how security logs can be used to identify unsuccessful sign-in attempts and support troubleshooting.

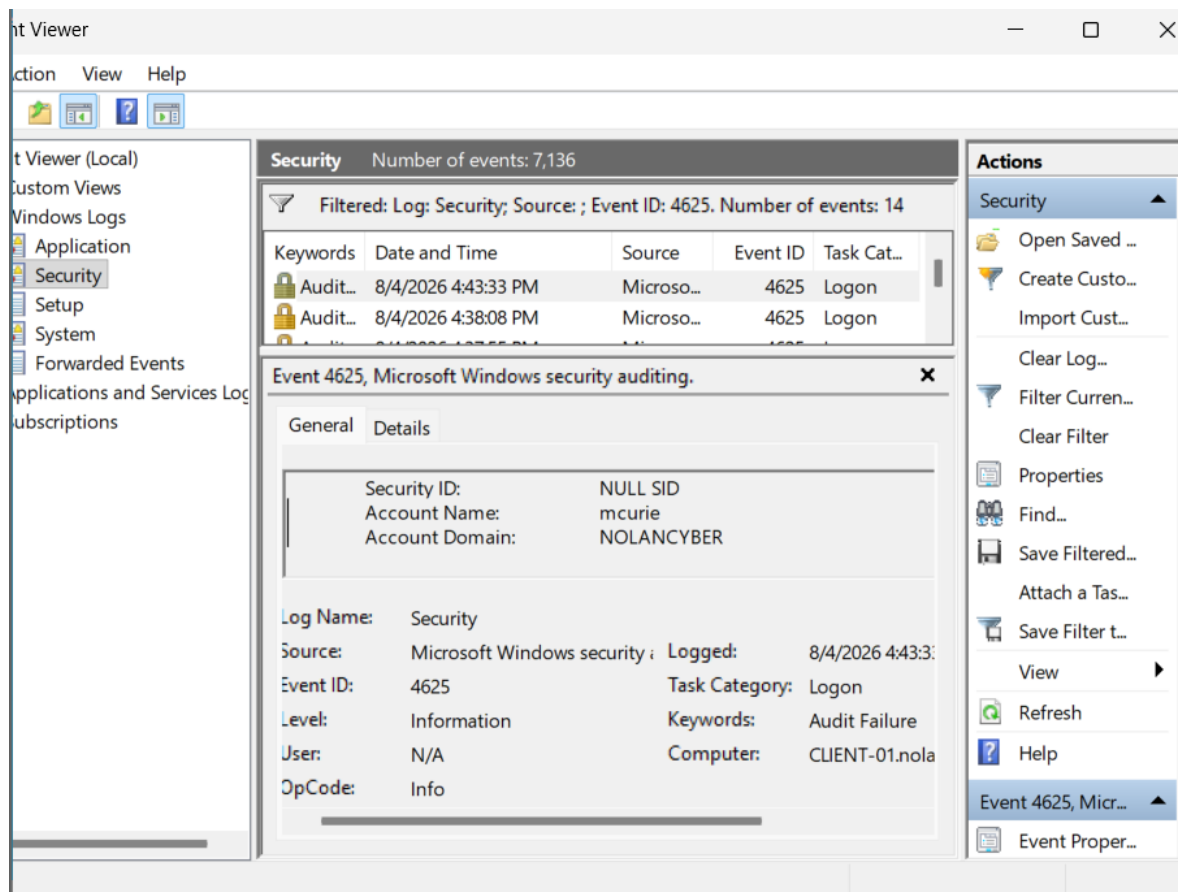


Figure 14. Event ID 4625 showing a failed authentication attempt.

11. DHCP Configuration

I installed and configured the DHCP Server role on DC01 to provide automatic IP addressing for domain workstations. I created an IPv4 scope using the address range 192.168.1.100–192.168.1.200 and configured the scope to provide DC01 (192.168.1.10) as the DNS server. After switching CLIENT-01 from a static IP address to DHCP, I verified that the client received an address from the configured scope and was using the correct DHCP and DNS servers.

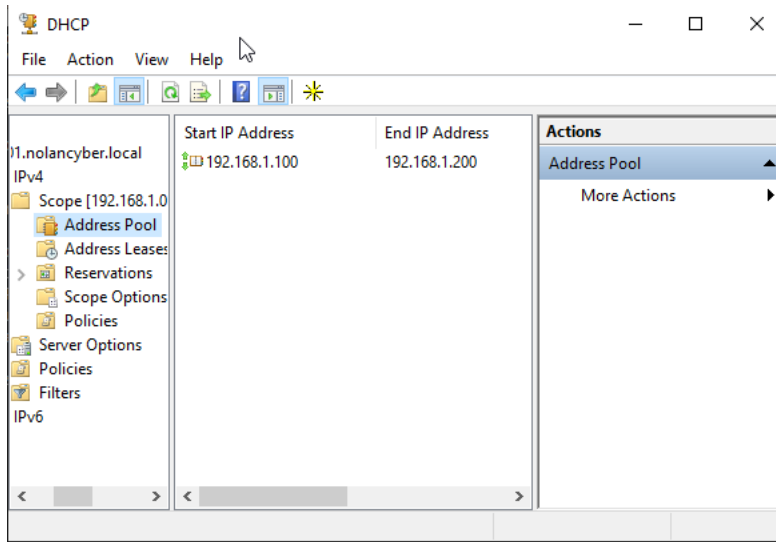


Figure 15. DHCP IPv4 scope configured on DC01 with an address pool of 192.168.1.100–192.168.1.200 for domain clients.

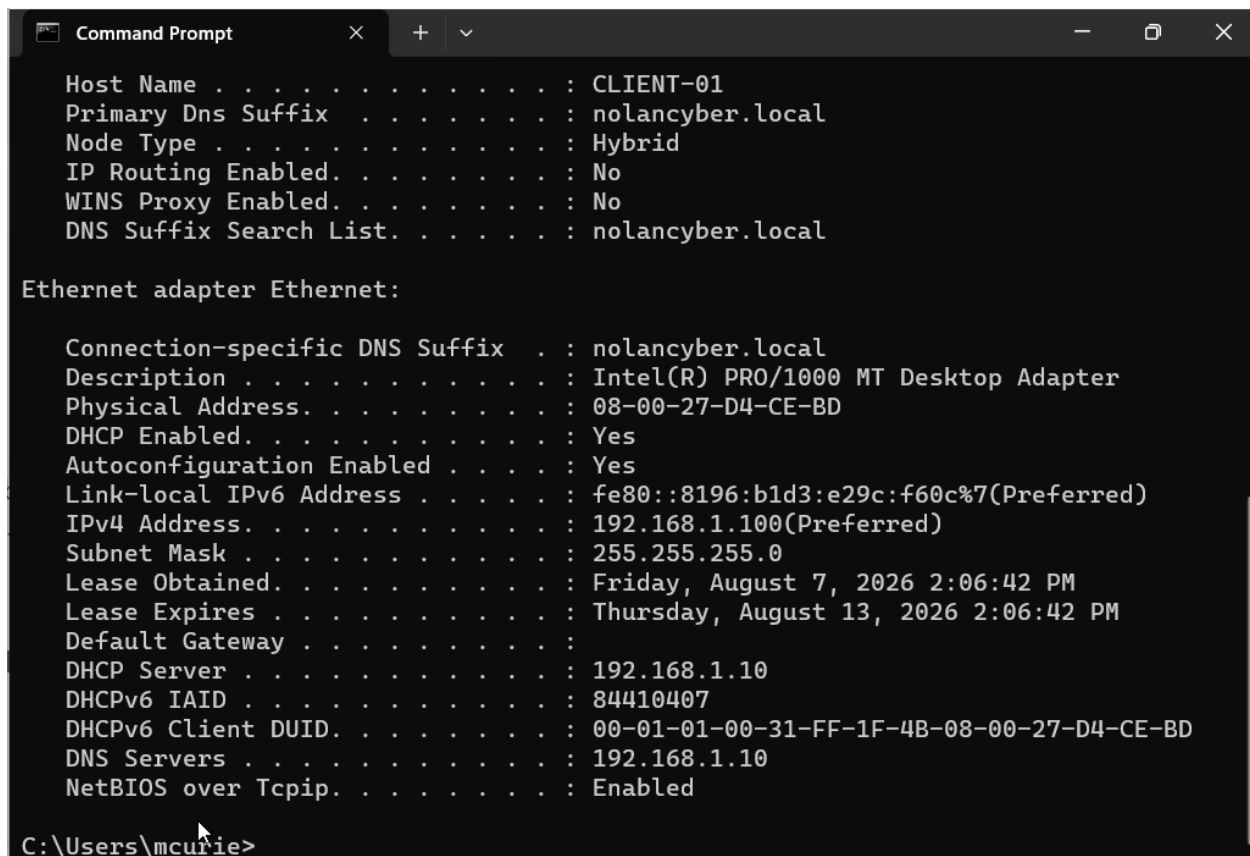


Figure 16. CLIENT-01 successfully receiving its network configuration from the DHCP server on DC01.

12. Automated Network Drive Mapping

I configured Group Policy Preferences to automatically map the NolanShares network share as the S: drive for domain users. This allows users to access the shared resource directly through File Explorer without manually entering the network path. The mapped drive provides access to the share, while the NTFS and share permissions configured earlier still determine which folders each user can access

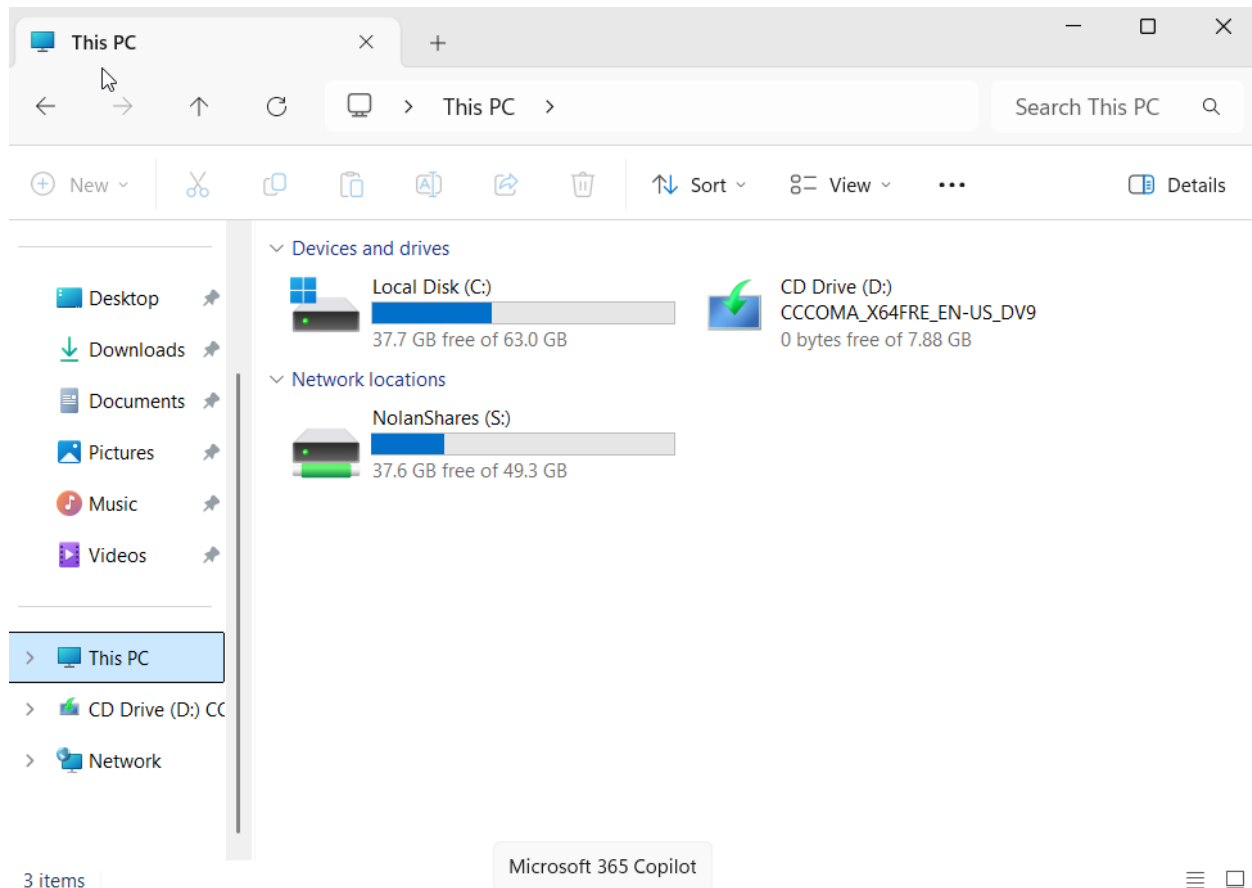


Figure 17. NolanShares automatically mapped as the S: drive on CLIENT-01 through Group Policy.

13. PowerShell Access Restriction

I created a Group Policy to restrict PowerShell access for standard departmental users while allowing members of the IT Administrators group to retain access. The policy was linked to the Departments OU, with the IT Administrators group excluded from applying the restriction. I tested the policy using both a standard Finance account and an IT account to confirm that PowerShell was blocked for the standard user but remained available to IT.

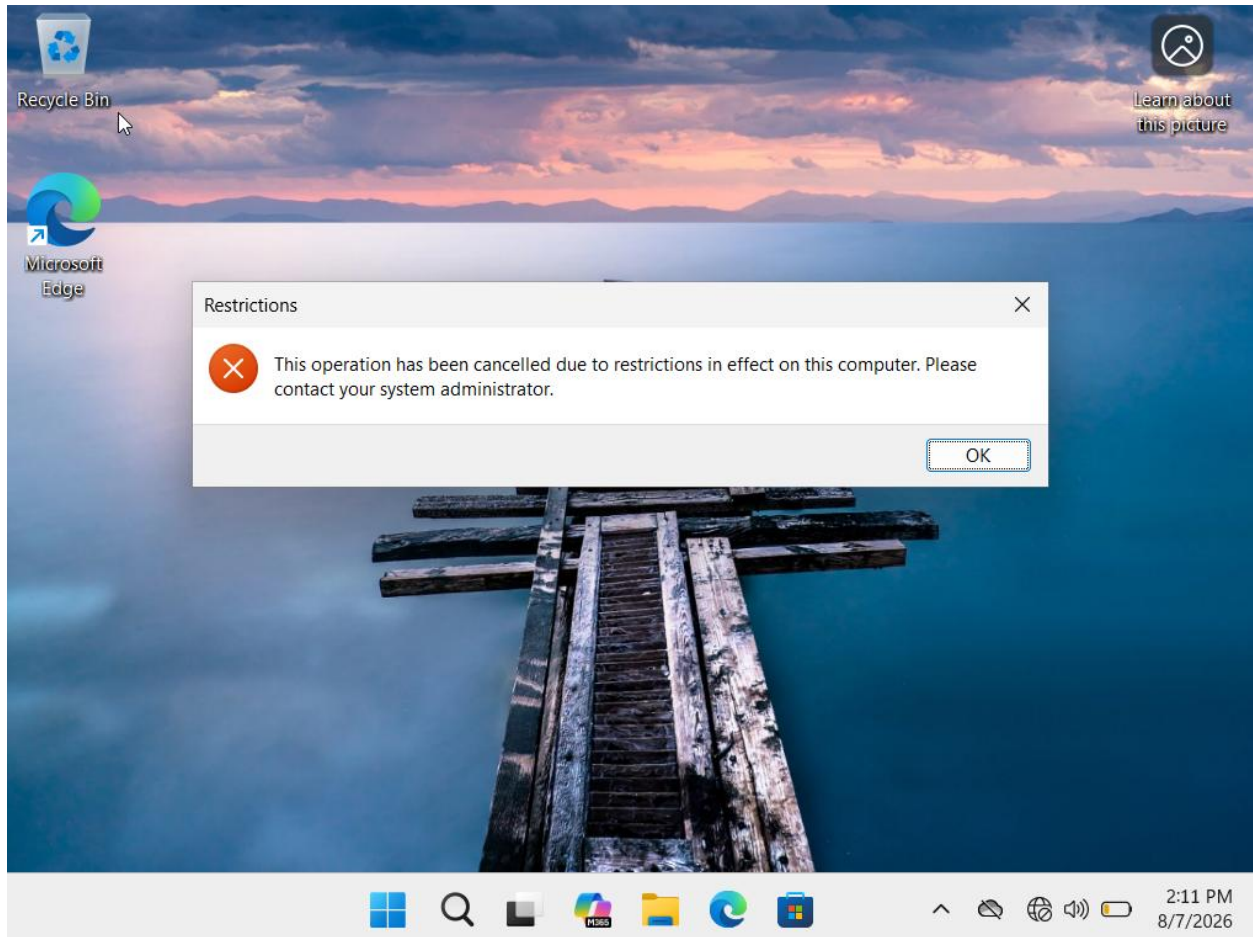


Figure 18. PowerShell access blocked for a standard Finance user through Group Policy.

14. Backup and Disaster Recovery

I installed Windows Server Backup and added a separate 25 GB virtual disk to DC01 for backup storage. The disk was initialized, formatted with NTFS, and configured as the AD Backup (B:) volume. I then performed a System State backup to protect critical domain controller components, including Active Directory and SYSVOL. This provides a recovery point that could be used if important server or directory data were lost or damaged.

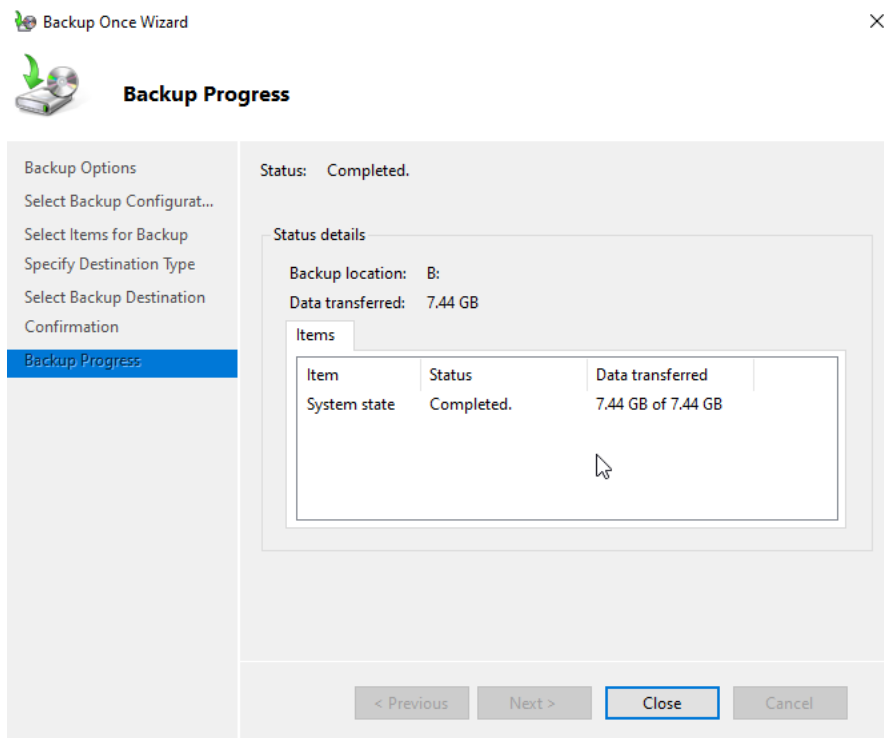


Figure 19. Successful System State backup of DC01 created with Windows Server Backup and stored on the dedicated AD Backup volume.

15. Troubleshooting

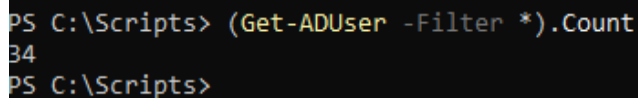
Issue	Resolution
PowerShell script errors	Corrected cmdlet parameter names and syntax.
CSV Import	Updated CSV formatting and verified OU paths and security groups.
Event Viewer	Confirmed the correct computer was being checked for audit events.
Time Synchronization	Corrected VM time configuration so both systems recorded accurate timestamps.

16. Additional Validation and Testing

Bulk User Provisioning

After setting up my original users, I wanted to test the PowerShell script on a larger group. I created a CSV with 20 additional users from different departments and updated the script so it could place each user in the correct OU and security group. After running it, the total number of users in the domain increased from 14 to 34.

I ran into an issue because the Human Resources group name in my script did not exactly match the name in Active Directory. The users were still created, but I had to correct the group name and verify the memberships afterward. This helped me understand how small naming differences can affect an automation script and why it is important to check the results after running one.



```
PS C:\Scripts> (Get-ADUser -Filter *).Count
34
PS C:\Scripts>
```

Figure 20. PowerShell verification showing 34 domain user accounts after batch-provisioning 20 additional users from CSV.

DHCP Reservation

I added a DHCP reservation for CLIENT-01 using its MAC address. I reserved 192.168.1.25 for the client so DHCP can give it the same IP address instead of having to configure the address manually.

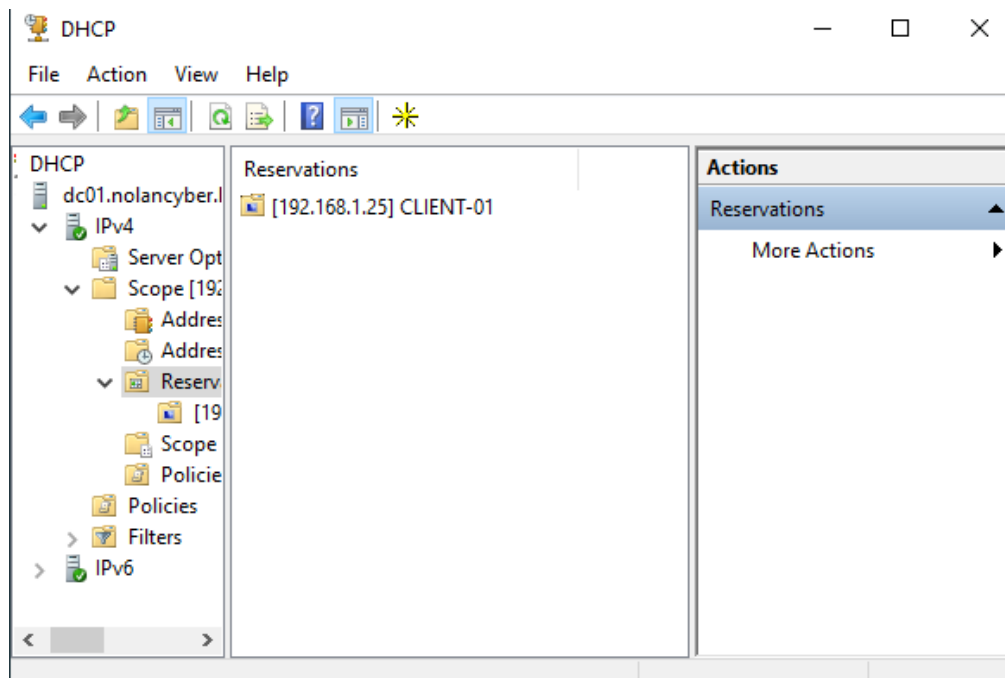


Figure 21. DHCP reservation configured for CLIENT-01 using its MAC address.

Access Testing

I tested the environment again using one of the new accounts created by the PowerShell script. I logged into CLIENT-01 as scooper, an Engineering user, and confirmed that the account could authenticate to the domain. When I opened the shared folders, the account could see the Engineering folder while folders for other departments were hidden. I also tried opening an administrative PowerShell window and confirmed that the standard user could not continue without administrator credentials.

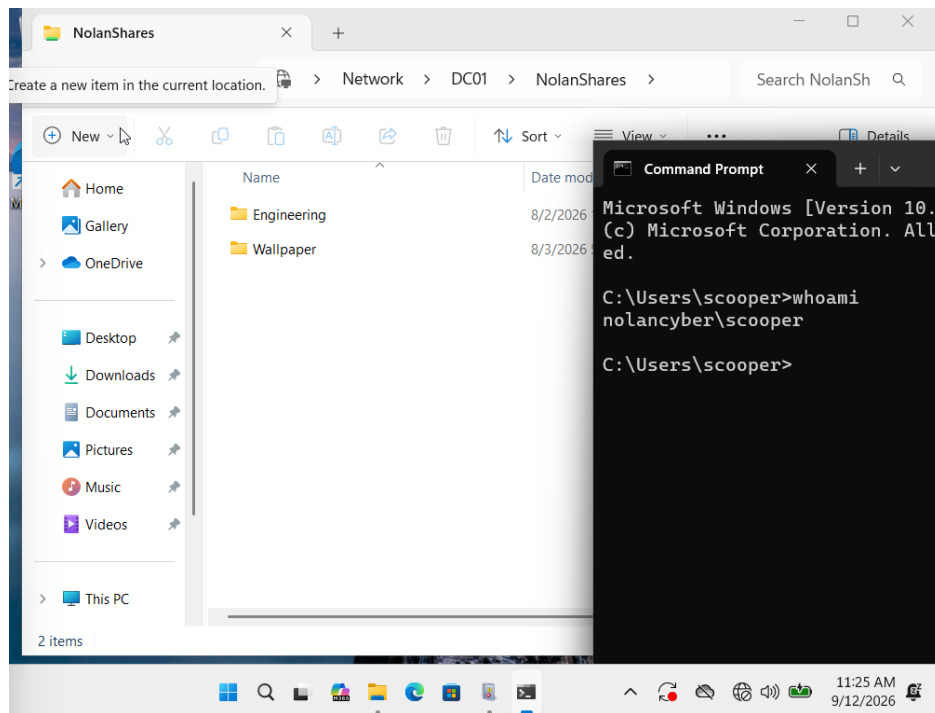


Figure 22. Share access test using the Engineering account scooper, showing the resources available to the user.

17. Skills Demonstrated

Core Technical Skills Demonstrated:

- Active Directory Domain Services (AD DS)
- Windows Server 2022 Administration
- Active Directory Users and Computers (ADUC)
- Group Policy Management
- User and Group Administration
- Organizational Unit (OU Design)
- User and Group Administration
- DNS Configuration
- Windows Client Domain Join
- NTFS and Share Permission Management
- Role Based Access Control (RBAC)
- PowerShell Scripting and Automation
- CSV-Based User Provisioning

- Windows Security Auditing
- Event View Log Analysis
- Oracle VirtualBox Administration
- Troubleshooting and System Validation

18. Conclusion

This project gave me hands-on experience building and managing a Windows Active Directory environment from the ground up. In addition to learning how to configure users, groups, policies, file permissions, and automation, I gained experience troubleshooting configuration issues and verifying that the environment worked as expected. Overall, the project strengthened my understanding of Windows Server administration and gave me practical experience with many of the tasks performed by IT administrators in business environments.

One of the biggest things I learned from this project was that troubleshooting is just as important as configuration. Several features didn't work correctly the first time, and working through those issues helped me understand Active Directory much better than simply following instructions would have.